Deep Learning-Based OCR for Medical Prescription Recognition

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GitHub Link: https://github.com/khanal123/OCR-for-Medical-Prescription-Recognition

Abstract

This paper presents a deep learning-based OCR system for extracting text from medical prescriptions. Models including CNN, CRNN, and attention-based architectures are explored. CRNN achieves better character-level accuracy, while attention models show faster convergence but instability. Evaluation is performed using CER and WER.

Keywords

OCR, deep learning, CNN, CRNN, attention, transformers, CER, WER

I. Introduction

Handwritten prescriptions are difficult to interpret and prone to errors. This work develops an OCR system using deep learning for accurate text extraction. Contributions include pipeline development, model comparison, and performance evaluation.

II. Related Work

CNN and CRNN models are widely used in OCR. Attention mechanisms improve sequence modeling. Transformer-based OCR models provide strong context understanding but require large datasets.

III. Methodology

A. Preprocessing

To prepare the dataset for OCR model training, several preprocessing steps were applied:

Image Preprocessing

• Images were converted to grayscale

• Images were resized to a fixed size (512 × 64)

• Pixel values were normalized between 0 and 1

Text Label Preprocessing

- Combining medicine name and dosage into a single sequence
- Example:
  - "Metformin 10mg Pantoprazole 40mg"
- Excluded the frequency field to:
  - Reduce sequence complexity
  - Improve model convergence

Sequence Handling

•Grouped labels per image to create one text sequence per prescription

•Removed very long sequences to ensure compatibility with the CTC loss used in CRNN

Dataset After Preprocessing

Total samples before filtering: 2,000

Total samples after filtering: 1,095

Text Length Statistics (Before Filtering)

Metric	**Value**
Mean	119.0360
Std	27.2362
Min	61.0000
25%	100.0000
50%	119.0000
75%	138.0000
Max	180.0000

Text Length Statistics (After Filtering)

Metric	**Value**
Mean	98.880365
Std	16.101975

Min	61.0000
25%	86.000000
50%	101.000000
75%	115.000000
Max	120.000000

Filtering significantly reduced sequence length variability, which improves training stability for sequence-based OCR models.

B. Model Architecture

Model 1:CRNN(CNN + RNN + CTC)

Architecture Components

- CNN (Convolutional Neural Network)
    - Extracts spatial features from images
- RNN (Recurrent Neural Network)
    - Captures sequential dependencies in both directions
- CTC (Connectionist Temporal Classification) Loss
    - Enables alignment-free training between input images and output text

Working Principle

The model converts an input image into a sequence of feature vectors, which are then processed by the BiLSTM and decoded into text using CTC.

Model 2: CNN +Attention + BiLSTM (Advanced Model)

Architecture Components

- CNN (Convolutional Neural Network)
    - Extracts spatial features from images
- BiLSTM (Bidirectional LSTM)
    - Learns sequential dependencies in text
- Attention Mechanism

    - Allows the model to focus on important parts of the image while predicting each character

- Decoder

    - Generates output text step-by-step

C. Training

Framework

The models were implemented using PyTorch which provides flexibility for building custom deep learning architectures for OCR tasks.

Training Configuration

The training setup was designed to ensure stable learning while handling sequence-based OCR data.

Common Hyperparameters

- Learning rate: 0.0001
- Batch size: 8
- Number of epochs: 10
- Optimizer: Adam

Model-Specific Training Details

1. CRNN (CNN + BiLSTM + CTC)

- Loss function:

    - CTC Loss (Connectionist Temporal Classification)
    - Allows alignment-free training between input images and output sequences

- Training Characteristics:

    - Handles variable-length sequences
    - No need for character-level alignment
    - Outputs probability distributions over characters per time step

- Additional Techniques:

  - The model was trained on a CPU due to compatibility limitations with the MPS backend
  - Gradient clipping was applied to stabilize training
  - Batches with incompatible sequence lengths were skipped

2. CNN +Attention+ BiLSTM Model

- Loss function:

  - Cross-Entropy Loss (with padding ignored)
  - Suitable for step-by-step character prediction

- Training Characteristics:

  - Uses teacher forcing during training
  - Predicts characters sequentially using attention
  - Requires properly padded sequences

- Additional Techniques:

  - The model was trained on a CPU due to compatibility limitations with the MPS backend
  - Gradient clipping was applied to stabilize training
  - Batches with incompatible sequence lengths were skipped

IV. Experimental Setup

A. Dataset Description and Exploratory Data Analysis(EDA)

The dataset used in this project is the Synthetic Medical Prescription OCR Dataset obtained from Kaggle. It is designed for training Optical Character Recognition (OCR) systems to extract structured medical information from prescription images.

The dataset contains:

- Prescription images

- Associated structured labels:

  - medicine name,
  - dosage
  - frequency

Each prescription image contains multiple medication entries.

Dataset Statistics

Total records: 13,694

Unique images: 2,000

Unique medicines: 10

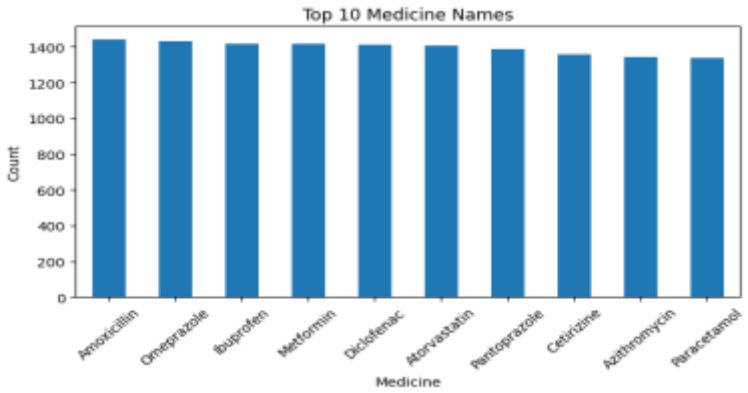
Unique dosages: 5

Missing values: None

Column	Missing Values
Image	0
Medicine	0
Dosage	0
Frequency	0

Exploratory Data Analysis (EDA)

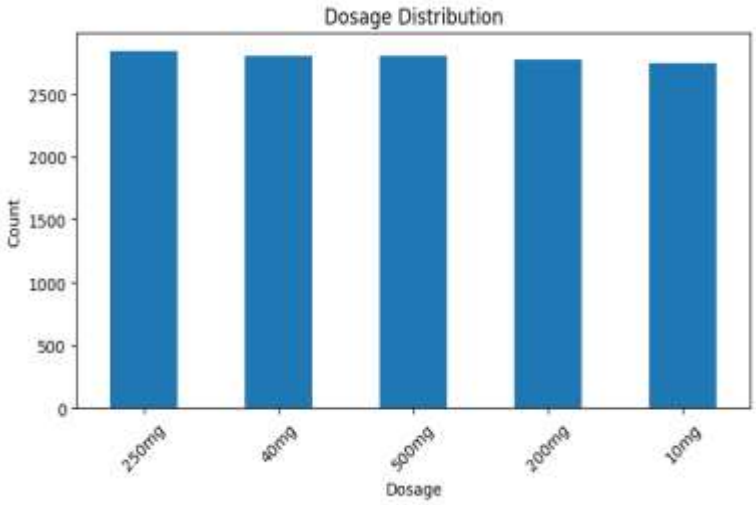
1. Most Frequent Medicines



The dataset contains 10 unique medicines, all of which appear with nearly equal frequency.

This indicates a balanced dataset, which is beneficial for model training and prevents bias towards specific medicines.

2. Dosage Distribution

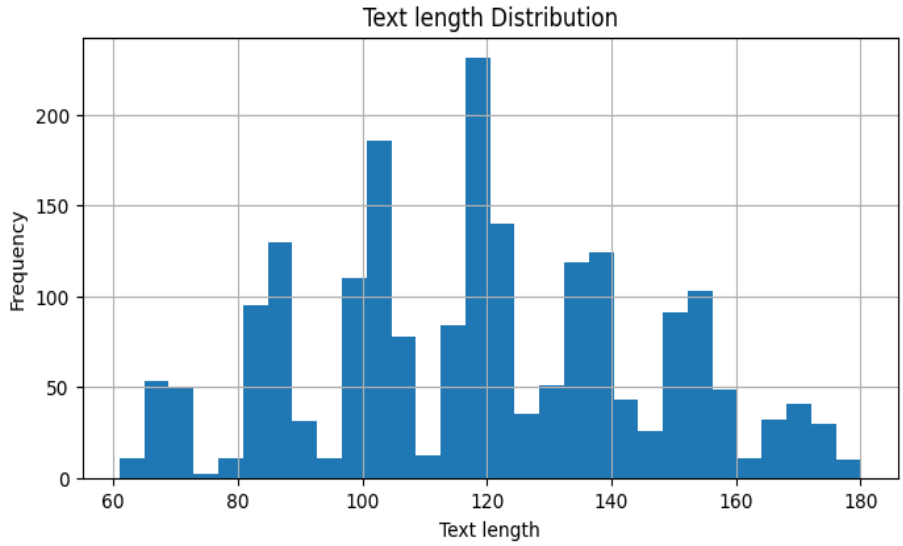


The dosage values include:

- **10mg, 40mg, 200mg, 250mg, 500mg**

Each dosage appears in roughly equal proportions, further confirming that the dataset is well-balanced across dosage classes.

3. Text Length Distribution



The text length distribution shows:

- Most sequences fall between 80 and 140 characters
- A peak around 100-120 characters
- Some longer sequences (up to 180) before filtering

After filtering:

- Maximum length reduced to 120
- Distribution becomes tighter and more uniform

This helps:

- Improve CTC alignment
- Reduce training complexity
- Prevent long-sequence learning issues

Summary of Findings

From the eda analysis, we can see that the dataset is clean and well-balanced, there were no missing values and Medicine and dosage distributions are uniform. Furthermore, text sequences are moderately long but manageable after filtering and preprocessing significantly improved data consistency and model compatibility. Overall, the dataset is well-suited for modern deep learning OCR architectures, enabling effective learning of both spatial and sequential text patterns.

B. Metrics

The models were evaluated using:

- Character Error Rate (CER)
    - Measures the percentage of incorrectly predicted characters
    - Based on edit distance between predicted and ground truth text
- Word Error Rate (WER)
    - Measures errors at the word level
    - More strict than CER, as one incorrect character can affect an entire word

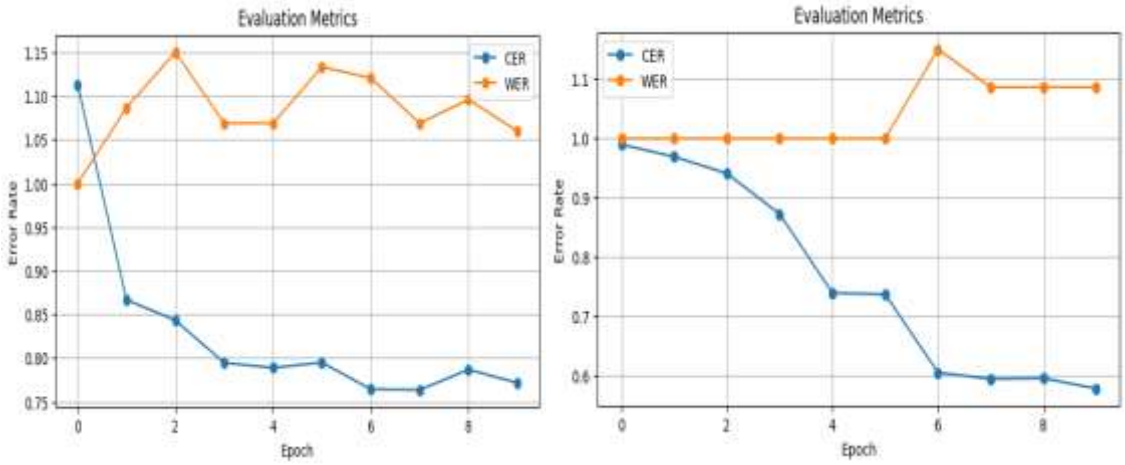
These metrics measure how accurately the predicted text matches the ground truth.

C. Implementation

The models were developed using PyTorch, a flexible deep learning framework that supports the design of custom architectures for OCR applications. Its dynamic computation graph simplifies experimentation and debugging, while features such as automatic differentiation and efficient tensor operations make it effective for training and optimizing sequence-based models like CRNN and attention-based networks.

V. Results

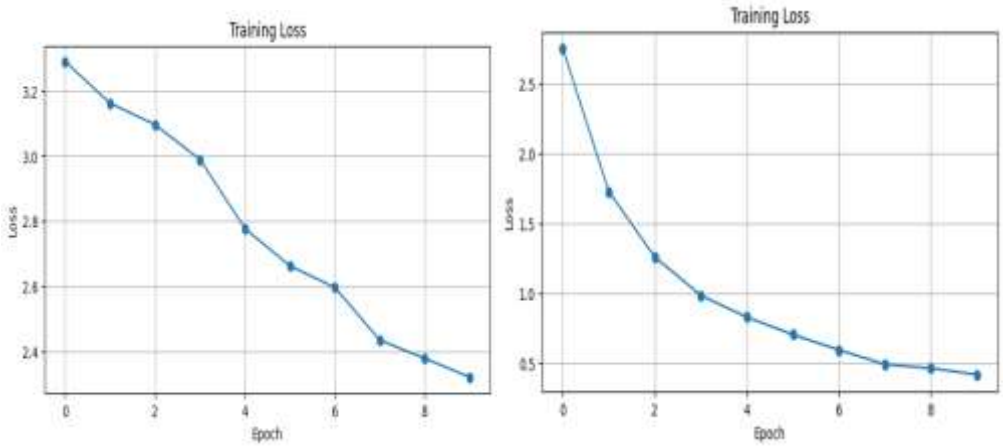
A. Main Results



The CRNN model achieves a lower final CER of approximately 0.57 while the CNN + Attention model reaches around 0.77. This indicates that CRNN is more effective in recognizing individual characters and provides better character-level accuracy.

Both models exhibit higher WER values, often exceeding 1.0, which indicates difficulty in predicting complete words correctly. However, the CRNN model demonstrates relatively more stable behavior, while the CNN + Attention model shows greater fluctuations and instability. This suggests that CRNN is better at maintaining sequence consistency compared to the attention-based model.

B. Analysis



The CNN + BiLSTM + Attention model shows a faster and more significant decrease in training loss, reducing from approximately 2.74 to 0.42 over 10 epochs. In contrast, the CRNN model demonstrates a slower decline in loss, from around 3.29 to 2.32. Although, the

CNN model converges faster, a lower loss does not necessarily indicate better OCR performance.

VI. Discussion

The CRNN model achieves better performance because it uses CTC loss, which allows effective alignment between image features and text sequences without needing explicit character segmentation. This makes the model more adaptable to differences in sequence length and spacing. On the other hand, the attention-based model faces challenges with repetition and longer sequences, as it depends on accurately learning alignments between image regions and characters. When this alignment is not properly learned, the model often produces repetitive or inconsistent predictions, particularly for more complex prescription texts.

VII. Conclusion

This project implemented an OCR system for medical prescription recognition using two deep learning approaches: a CRNN model and a CNN + BiLSTM + Attention model. Both models were trained to convert prescription images into text sequences and evaluated using Character Error Rate (CER) and Word Error Rate (WER).

The results show that both models exhibit learning behavior, as indicated by decreasing training loss and improvement in CER over epochs. However, the CRNN model demonstrates better performance overall, achieving lower CER and more stable predictions. In contrast, the CNN + Attention model, although showing faster loss convergence, struggles with sequence alignment and produces repetitive outputs.

VIII. Future Work

Implement transformers, larger datasets, better augmentation, and sequence alignment improvements.

AUTHORS

Suman Khanal

Designed and implemented the CNN based OCR model, performed data preprocessing and contributed to model training and evaluation

Baradan Mainali

Designed and implemented the CRNN-based OCR model, handled sequence modeling and contributed to performance analysis and comparison

References

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